Data-efficient surrogates for vocal-fold models: a comparison of low-N regression strategies

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Abstract

Physically based vocal-fold models predict phonatory outputs from muscle activations and subglottal pressure but are expensive to evaluate, so machine-learning surrogates are used in their place. Surrogate accuracy, however, collapses when only a handful of simulations are available. Low sample use cases are especially relevant in subject-specific clinical use, where 10 to 100 target samples may be all that can be obtained. Transfer learning was the natural remedy: train a surrogate on a cheap source model (a body-cover model, BCM) and adapt it to an expensive target with a few target simulations. This has been shown to work between closely related lumped models, for example BCM to Triangular Body Cover Model (TBCM). Here we ask whether it extends to a higher-fidelity target: a structurally distinct finite-element Beam and Membrane (BM) model. It does not. Used on BM without adaptation the BCM source is far less accurate than predicting each output's mean (F_0 $R^2 = -30.9$), and the optimized transfer methods built on it improve on the same regressor fit to the target data alone by only $+0.03$ R^2 on average at $N = 50$, against a gain of $+0.11$ on the aligned target. We show that a no-source alternative closes this gap. TabPFN, a tabular transformer pretrained once and then frozen, fits a task in-context with no source model and no training. On the far BM target it is the most accurate method from $N = 50$ upward (F_0 $R^2 = 0.89$ at $N = 50$) and it reconstructs the full ($a_{CT} \times a_{TA}$) activation map with lower image-level error than transfer at every sample size tested. On the aligned TBCM target, where the source is a genuinely good prior, optimized transfer remains ahead at the smallest sample sizes (F_0 $R^2 = 0.93$ versus 0.78 at $N = 10$) and TabPFN overtakes it from $N = 100$. The practical consequence is that the no-source model is the only one of the seven that is strong on both targets without needing a source to be available or well matched, whereas transfer is competitive only when the source is already close. We limit our conclusions to the methods, targets, and forward setting studied.

1 Introduction

Physically based models of the vocal folds, such as lumped body-cover models and higher-fidelity continuum (finite-element) models, predict phonatory outputs from muscle activations and subglottal pressure, but are costly to evaluate at scale. Prior work showed that machine-learning surrogates can replace these simulations for the forward problem, yet their accuracy degrades sharply when few training simulations are available. That degradation motivated transfer learning: train a surrogate on a cheap source model (the body-cover model, BCM) and adapt it to an expensive target with a small number of target simulations. In practice, obtaining data from complicated models is costly, so being able to circumvent that with a machine learning surrogate is ideal. This transfer has been shown to succeed between closely related lumped models (BCM to TBCM). [TODO: insert citation]

That prior result raises the question this paper investigates: does the transfer-learning approach extend from closely related lumped models to a higher-fidelity target? A cheap source helps only while it remains a good prior for the target, and a finite-element model is both more detailed than the lumped source and structurally unlike it, so whether the established approach carries over is not obvious. Our central question, and its fallback, is therefore:

> Does data-efficient, subject-specific surrogate modeling, shown to transfer between lumped vocal-fold models (BCM to TBCM), extend to a higher-fidelity target (BCM to BM)? And where transfer does not carry over, can a no-source pretrained model serve in its place?

We approach this in two steps. **Aim 1** (data efficiency, Sec. 3.1) measures three regressor families, polynomial, random forest, and neural network, each fit two ways: on the target data alone, to establish a baseline, and with optimized transfer learning from the source. **Aim 2** (fidelity extension, Sec. 3.2) tests whether the transfer benefit that holds on the aligned lumped target survives the move to the structurally distinct, higher-fidelity BM target. Throughout we also evaluate TabPFN [1], a transformer pretrained once on millions of synthetic tabular problems and then frozen, which fits a new task in-context in a single forward pass with no per-task training and no source model. We let the results, not the premise, identify the strongest method, and we bound our conclusions to the methods and targets evaluated.

2 Methods

2.1 Physical vocal-fold models and data

All models share the inputs (cricothyroid, thyroarytenoid, and lateral cricoarytenoid activations a_{CT}, a_{TA}, a_{LCA}, and subglottal pressure P_s) and the base outputs F_0 and SPL; the richer physiological outputs (Sec. 3.3) are added for the targets. All surrogates are fit on the fixed input schema $[a_{CT}, a_{TA}, P_s]$. The three models span a deliberate range of source–target fidelity:

- **Body-Cover Model (BCM), source.** A lumped-element model and the cheap source (360,750 samples) for all transfer methods.
- **Triangular Body-Cover Model (TBCM), aligned target.** Lumped-element, the same modeling family as BCM, so the source–target gap is small. 43,102 samples, drawn on a 40×40 ($a_{CT} \times a_{TA}$) grid across 31 lung-pressure levels, providing dense ODE ground-truth activation maps (Sec. 3.2). This is the near target for Aim 2.
- **Beam and Membrane (BM), far target.** A finite-element continuum model; the expensive, structurally distinct target. Relative to the lumped BCM source it is the far target, where the low-N question is sharpest and the fidelity gap is largest. 4,647 phonating samples (filtered ACFL > 30 from 5000 simulations).

The BCM to BM gap is not merely one of accuracy but of representation: BM resolves distributed tissue displacement fields where BCM lumps them into a few masses, so quantities such as AC flow can differ in definition and scale between the two (Sec. 3.3). [TODO: add model equations and refs; cite Parra TBCM and the BM solver; state the BCM to BM structural differences precisely.]

2.2 Low-N regression strategies

We evaluate three standard regressor families, each fit two ways, plus a pretrained no-source model, giving seven configurations under one protocol.

Regressors fit on the target alone (baseline). To measure what transfer learning actually contributes, we first fit each regressor family on the target data by itself, with no source model. This is not a proposed method but a reference point: if a transfer method cannot beat the same regressor trained on the target data alone, the source has added nothing. Every transfer and no-source result below is read against this reference.

- **Polynomial (PR).** Degree-3 polynomial features on standardised inputs, followed by ridge regression whose penalty is selected per target by leave-one-out generalised cross-validation over $\alpha \in [10^{-3}, 10^{4}]$ (15 log-spaced values). The ridge penalty is what makes the degree-3 basis usable at $N = 10$, where the feature count exceeds the sample count.
- **Random forest (RF).** 300 trees throughout, with depth and leaf size scaled to the sample count so that capacity tracks the data available: $N < 50$: depth 3, min leaf 1; $N < 250$: depth 6, min leaf 2; $N < 1000$: depth 10, min leaf 2; $N \geq 1000$: depth unlimited, min leaf 1. Tree count does not drive overfitting in a bagged ensemble, so it is held fixed and capacity is controlled by depth.
- **Neural network (NN).** A feed-forward network with two hidden layers of 64 ReLU units and one linear output per target, trained full-batch with Adam (learning rate 0.003, weight decay 0.0001) for 800 epochs on standardised inputs and outputs. The source-pretrained copy used by the transfer variant is trained on a 20,000-row subsample of the source for 300 mini-batch epochs at learning rate 0.001.

Per-domain standardisation is used throughout: each domain fits its own input scaler and its own per-output scalers, and no scaler is ever shared across domains.

Optimized transfer learning (per family). Each family also appears as an optimized transfer method, and the procedure is identical for all three. A source model of that family is pretrained on BCM and combined with three sub-models fit on the N target rows:

1. **target-only**: the family fit on (X, Y) (this is exactly the baseline above);
2. **residual correction**: the family fit on $(X,\ Y - \hat{Y}_{\text{src}})$, predicting $\hat{Y}_{\text{src}} + f(X)$;
3. **feature augmentation**: the family fit on $([X, \hat{Y}_{\text{src}}],\ Y)$, taking the source prediction as an extra input.

The three are blended per output by non-negative weights that sum to one. The weights are fit on *out-of-fold* sub-model predictions ($K = \min(5, N)$-fold), not on the training rows themselves. This matters: if the weights are fit in-sample, the target-only sub-model can win weight purely by memorising the N training rows, which would bias the transfer method toward the baseline and make "transfer ≈ baseline" an artifact of the blending rather than a finding. Out-of-fold stacking gives the source-dependent sub-models their fair chance to earn weight.

TabPFN (no source). A pretrained tabular transformer applied with per-output standardisation, one single-output head per output. No training beyond the in-context fit, and no source model.

Evaluation. For each training size $N \in \{10, 20, 50, 100, 200, 500\}$, every method trains on the same N target rows and is evaluated on the same disjoint held-out pool of up to 1000 rows, reporting mean $\pm$ standard deviation over 5 seeds. We report accuracy (R^2) and a range-normalized RMSE,

$$\text{nRMSE} = \frac{\sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}}{\max_i y_i - \min_i y_i}, \tag{1}$$

rather than MAPE, because SPL, collision pressure, and CPP cross zero, which makes a percentage error ill-defined. Reporting a normalized RMSE alongside R^2 also allows direct comparison against prior surrogate work that reports error rather than variance explained. All methods share the same splits and seeds, so differences reflect the method rather than the data draw.

3 Results

3.1 Aim 1, data efficiency: accuracy versus training size

On the far (BM) target, TabPFN attains the highest R^2 and the lowest normalized RMSE in 52 of 60 (training size $\times$ output $\times$ metric) comparisons, and is the best method at every training size from $N = 50$ upward (Fig. 1, Table 1). At the two smallest sizes it is matched, on F_0, by the neural network fit on the target alone (0.74 versus 0.71 at $N = 10$); with 5 seeds these two are within each other's spread, so we read them as tied rather than ordered. The cheap BCM source is a poor prior for the finite-element BM model. Used on BM without adaptation it reaches an F_0 R^2 of -30.9, far less accurate than simply predicting each output's average value, so the transfer methods have nothing useful to build on and gain little over fitting the same regressor on the target data alone: averaged over the three families and five outputs, optimized transfer improves on the baseline by +0.03 R^2 at $N = 50$ and +0.01 at $N = 500$ (Table 1). TabPFN, which needs no source at all, is ahead of both by +0.06 R^2 on average at $N = 50$. The richer physiological outputs (Sec. 3.3) follow the same ordering. As N grows all methods converge toward high accuracy, and TabPFN stays in front from $N = 50$ onward.

Table 1: Aim 1. Accuracy (R^2, upper block) and range-normalised RMSE (lower block) on the far BM target at $N = 50$ target simulations, mean over 5 seeds. Best per row in bold. The *target alone* columns are each regressor family fit on the N target rows with no source; *optimized transfer* adds the BCM source through the blended procedure of Sec. 2.2.

	Fit on target alone			Optimized transfer			No source
Output	PR	RF	NN	PR	RF	NN	TabPFN
R^2 *(higher is better)*							
F0	0.85	0.60	0.79	0.85	0.77	0.85	**0.89**
SPL	0.72	0.69	0.68	0.72	0.67	0.72	**0.81**
ACFL	0.89	0.78	0.85	0.89	0.79	0.87	**0.92**
PC	0.57	0.60	0.36	0.61	0.58	0.47	**0.71**
CPP	0.36	0.40	-0.04	0.41	0.38	0.03	**0.47**
nRMSE (lower is better)							
F0	0.09	0.14	0.10	0.09	0.11	0.09	**0.08**
SPL	0.07	0.07	0.07	0.07	0.07	0.06	**0.05**
ACFL	0.07	0.09	0.08	0.07	0.09	0.07	**0.06**
PC	0.15	0.14	0.18	0.14	0.15	0.16	**0.12**
CPP	0.16	0.15	0.20	0.15	0.16	0.19	**0.15**

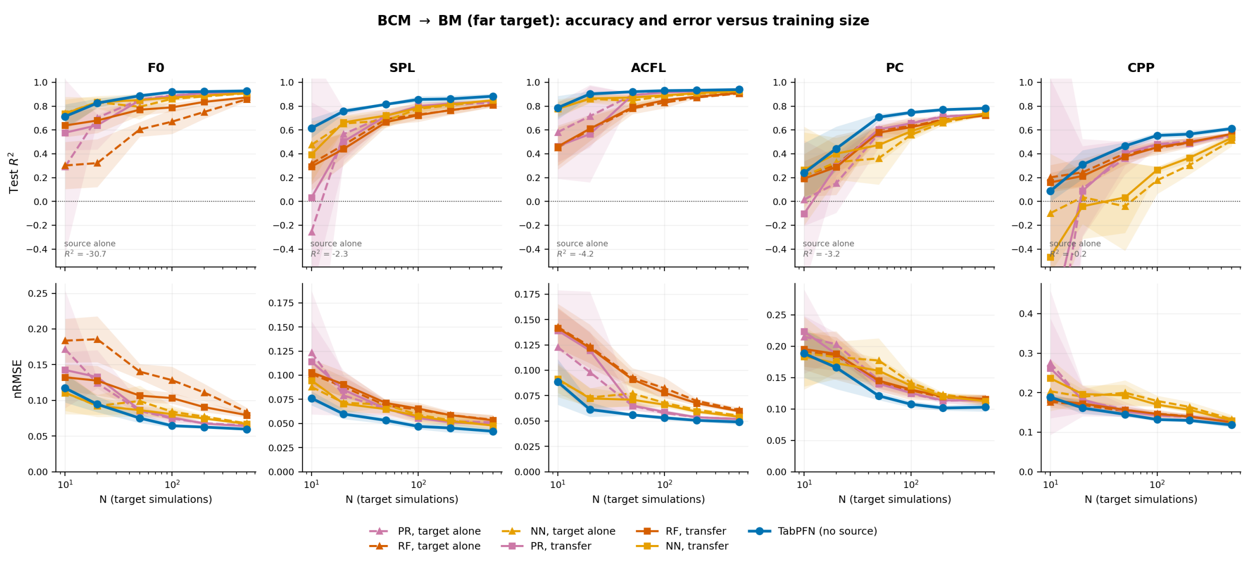

Figure 1: **Aim 1, data efficiency.** Accuracy and error versus training size on the far BM target (all five outputs): R^2 (top) and normalized RMSE (bottom) versus N for the regressors fit on the target alone (dashed, our baseline), the optimized transfer methods (solid), and TabPFN (thick blue), mean ± s.d. over 5 seeds. Colour encodes the regressor family and line style the variant, so each baseline/transfer pair can be read directly: the two curves of a colour nearly coincide on this target, which is the visual form of the result that the BCM source adds little. The source-alone R^2, annotated per panel, falls far below the axis and illustrates the poor BCM prior on this target.

3.2 Aim 2, model complexity: when does transfer help?

Whether the source helps at all is judged against the baseline, each regressor fit on the target data alone. On the aligned lumped target (TBCM, the same family as the BCM source), the source is already a strong prior, by itself reaching an F_0 R^2 of 0.86 with no target data, and optimized transfer clears the baseline, raising the F_0 R^2 from 0.94 to 0.97 at $N = 50$. On the far continuum target (BM), the same source reaches an F_0 R^2 of -30.9, less accurate than guessing each output's average value, and transfer does not clear the baseline at all: the best target-alone family reaches an F_0 R^2 of 0.854 and the best transfer variant 0.851, a difference of -0.003. The source is not merely unhelpful here, it is dead weight. Transfer's payoff therefore tracks how close the source is to the target, exactly the principle it rests on.

TabPFN, which uses no source, is the strongest method on BM from $N = 50$ upward and on TBCM from $N = 100$ upward, with an F_0 R^2 of 0.96 on TBCM and 0.89 on BM at $N = 50$. It is not uniformly best: on the aligned target at the smallest sample sizes, optimized transfer is clearly ahead of it (0.93 versus 0.78 at $N = 10$). That exception is not a counterexample to the principle but an instance of it. Transfer wins exactly where a well-aligned source is available and target data is scarcest, which is the regime transfer was designed for; the moment the source stops being a good prior, as on BM, that advantage disappears entirely. What the no-source model buys is not universal superiority but insensitivity to the fidelity gap. Its worst shortfall against the best method in any target-and-sample-size combination we tested is 0.15 R^2, and it occurs on the aligned target where a good source exists; the transfer family's worst shortfall is 0.22 R^2, and it occurs on the far target where no good source is available and none can be substituted.

Table 2: Aim 2. Source–target fidelity. F_0 accuracy (R^2) at $N = 50$ as the target moves from an aligned lumped model (TBCM) to a structurally distinct continuum model (BM). *Baseline* is the best of the three regressor families fit on the target alone; *opt. transfer* is the best of the three transfer variants. Transfer is only worth its source when it beats the baseline column. A negative source-alone value means the unadapted BCM source is less accurate than predicting the mean output.

Target (fidelity)	Source alone	Baseline	Opt. transfer	TabPFN
TBCM (aligned lumped)	0.86	0.94	**0.97**	0.96
BM (far continuum)	-30.9	0.85	0.85	**0.89**

Beyond point accuracy, a useful surrogate must reconstruct the physiological structure of the ($a_{CT} \times a_{TA}$) control space. On TBCM, which carries a dense ODE ground-truth map on the complete 40×40 grid, TabPFN reconstructs the full motor-control surface to an image-level R^2 above 0.9 from $N = 25$ samples and converges to the ground truth by $N = 1000$. It has lower image-level error than optimized transfer at 6 of the 9 training sizes tested ($N \geq 50$) (Fig. 2); below that, transfer reconstructs the surface better, mirroring the point-accuracy ordering on this aligned target. On BM the separation is unambiguous: against a dense full-data reference TabPFN has the lower image-level error at every training size, reaching an image-level R^2 above 0.9 by $N \approx 25$, while the transfer surface stays blocky and dragged toward the misaligned source and never separates from the target-alone baseline.

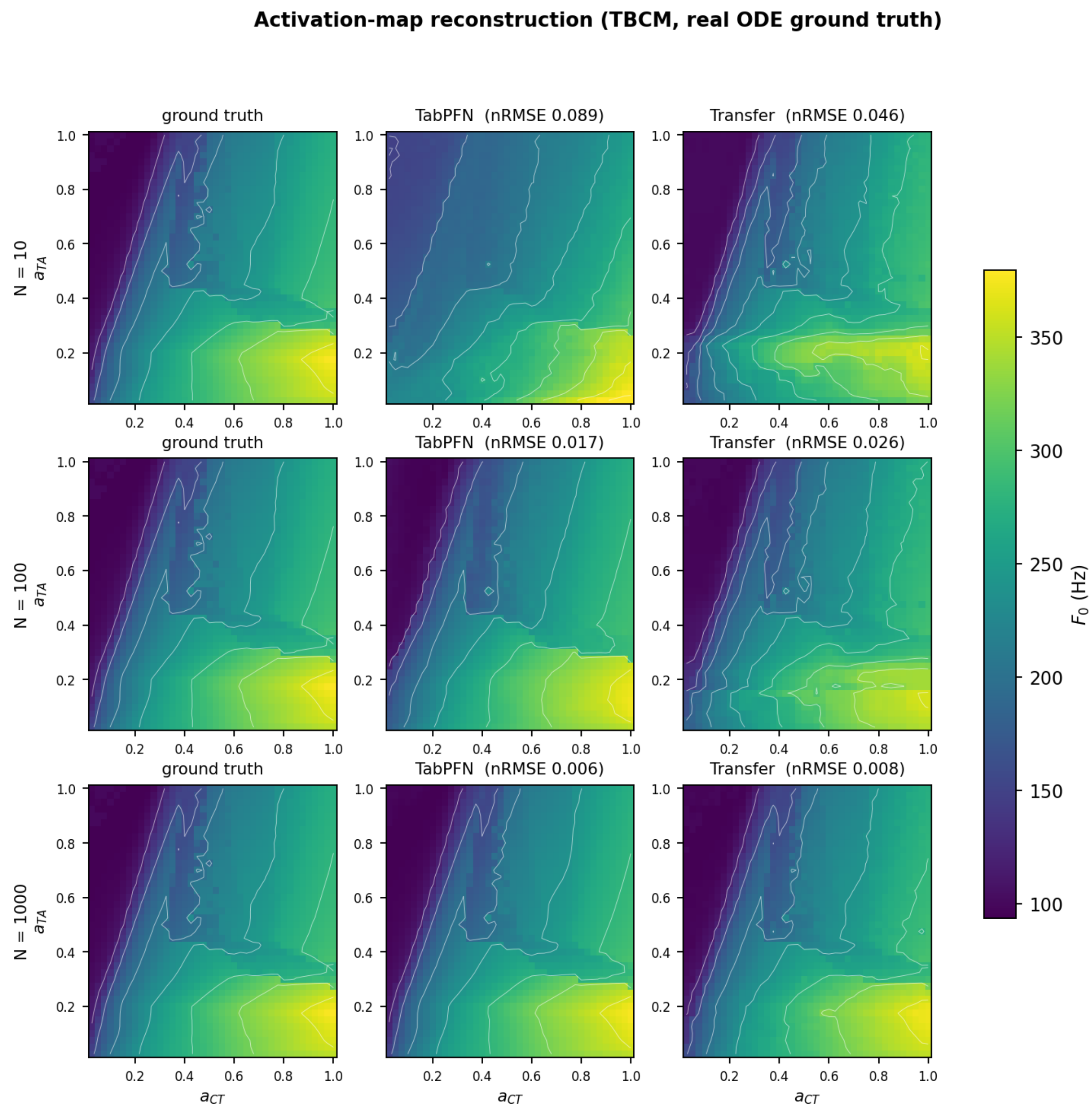

Figure 2: **Activation-map reconstruction (TBCM, real ODE ground truth).** F_0 motor-control map over (a_{CT}, a_{TA}) on the complete 40×40 grid at $P_L = 1250$ Pa. Columns: ODE ground truth, TabPFN, optimized transfer; rows: increasing N (10/100/1000). Image-level nRMSE over all 1600 cells is given per panel. TabPFN recovers the correct structure from small samples; the transfer surface is distorted at low N and sharpens only as N grows.

3.3 Aim 3, richer physiological outputs

F_0 and SPL are the conventional surrogate targets, but clinical use turns on quantities that describe tissue loading and voice quality. We therefore extend the BM target with three further outputs: AC glottal flow (ACFL), peak collision pressure (PC), and cepstral peak prominence (CPP). The BM dataset was regenerated row-aligned to the original 5000 operating points so that these outputs correspond to exactly the same simulations as the existing F_0 and SPL values.

The ordering of methods is unchanged on the richer outputs: at $N = 50$ TabPFN leads on all five of the five outputs, and the mean transfer gain over the target-alone baseline across ACFL, PC and CPP is +0.03 R^2 (Table 1). What does change is the difficulty ranking. ACFL is the easiest output for every method (TabPFN $R^2 = 0.92$ at $N = 50$), while CPP is by a clear margin the hardest (TabPFN $R^2 = 0.47$), with PC intermediate ($R^2 = 0.71$). CPP is a cepstral voice-quality measure computed from the radiated pressure waveform rather than a direct mechanical quantity, and it is the output least determined by the three input activations alone, which is the most likely reason no method reaches high accuracy on it at any N we tested. This is a limit of the input parameterisation rather than of any one regressor.

4 Discussion

The two aims resolve into one picture. On data efficiency, a pretrained tabular transformer is the strongest low-N surrogate among the methods tested on the far target, and is either the best or within 0.15 R^2 of the best on the aligned one. On model complexity, the benefit of transfer learning is governed by how close the source is to the target: it is large when the target is close to the cheap source (TBCM) and negligible when the target is a structurally distinct, higher-fidelity model (BM). Because TabPFN needs no source model or source data, it is insensitive to that fidelity gap, needing neither a well-aligned source nor the per-domain pretraining a source demands.

We state these conclusions narrowly. They hold for the seven configurations evaluated, the two targets studied (TBCM and BM), and the forward problem; they do not establish that TabPFN is optimal in general, nor do they speak to methods we did not test (for example, physics-informed networks) or to targets and output sets we did not examine.

In particular, our result should not be read as a claim that transfer learning requires structural similarity in general. It says that *this* source, under *these* transfer procedures, does not carry to *this* continuum target. Transfer has been applied successfully between models that are architecturally quite dissimilar, including from a triangular body-cover model to clinical measurements, where the source and target differ far more than BCM and BM do. [Emiro: please add the citation for the TBCM-to-clinical transfer work here, and correct this characterisation if it misstates the setup.] What our result isolates is narrower. At the level of the whole domain, the contrast is large and unambiguous: the unadapted BCM source reaches an F_0 R^2 of 0.86 on TBCM and -30.9 on BM, and mean transfer gain over the target-alone baseline follows it, +0.08 against +0.02. A source drawn from the same lumped-element family as the target is worth building on; one facing a structurally distinct continuum model is not.

We are deliberately not turning that into a per-output predictive rule, because our data cannot support one. Across the 7 (domain, output) pairs we measured, neither unadapted R^2 nor rank agreement with the target correlates significantly with realised transfer gain (Spearman -0.43, $p = 0.34$ and +0.64, $p = 0.12$ respectively). With seven pairs from two targets, these are descriptive, not inferential.

What the per-output numbers do suggest is a mechanism worth testing properly on more targets. R^2 against a target conflates two unlike failures: a miscalibrated source has the wrong scale but the right ordering, which residual correction and feature augmentation exist to undo, whereas a structurally disagreeing source ranks operating points wrongly and no downstream sub-model repairs that. BM's own F_0 is the illustration. It has the worst unadapted R^2 of any pair we measured, -30.9, and yet the largest low-N transfer gain on that target, +0.13, because its rank agreement with the source is also the highest on BM (Spearman 0.73 against 0.31 for CPP). Ranking candidate sources by rank agreement or by affine-corrected R^2, rather than by raw accuracy, is therefore a hypothesis this work motivates but does not establish.

The blend weights are consistent with that reading. Fitted out-of-fold, and never told which domain they are in, they place 58% of the weight on the source-dependent residual sub-model on TBCM, whereas on BM the weight sits on the target-only sub-model and its share rises from 0.37 at $N = 10$ to 0.49 at $N = 500$. The procedure ends up relying on the BM source less the more target data it has.

Transfer learning for a no-source model (future work). Because TabPFN wins at low N but has no weights to fine-tune, giving it a form of transfer is an open direction: rather than weight transfer, a linear combination of source-fit and target-fit in-context models may import source structure while preserving the no-training property. [TODO: report feasibility once explored.]

A property under missing data (future work). A pretrained in-context model can also be queried when parts of the input space are unobserved. In preliminary experiments, TabPFN reconstructs held-out regions of the control space and degrades gracefully as the missing fraction grows. Because this is not straightforwardly comparable across all methods, we note it here as a direction rather than a result.

5 Conclusion

Across a controlled comparison of three regressor families (each fit on the target alone and with transfer) and a pretrained tabular transformer, on targets spanning a range of source fidelity, two findings hold. First, a pretrained in-context model is the most accurate low-N surrogate on the structurally distinct target, with no source and no training, and is competitive on the aligned one, where optimized transfer leads at the smallest sample sizes. Second, the payoff of transfer learning is set by how close the source is to the target: substantial for a closely related lumped target, where it is the best method of all at the smallest sample sizes, and negligible for a far continuum target, where it does not improve on fitting the target data alone. A no-source in-context model sidesteps that dependence, which is what makes it the safer default when the source–target relationship is unknown. We present this as a finding about the methods and targets evaluated, not as a universal claim.

References

[1] N. Hollmann, S. Müller, K. Eggensperger, and F. Hutter, "TabPFN: A transformer that solves small tabular classification problems in a second," in *International Conference on Learning Representations (ICLR)*, 2023.